The Ladder

MIDDLE SCHOOL · RATIOS TO THE COORDINATE PLANE

LEARN

Two ratios are **equivalent** when one is the other scaled by the same factor top and bottom. To solve $\frac{a}{b} = \frac{c}{d}$ you may cross multiply, because multiplying both sides by bd clears both denominators at once: $ad = bc$.

Slope measures how steep a line is. Between any two points on it, $m = \frac{y_2 - y_1}{x_2 - x_1}$ — the rise over the run.

Try it

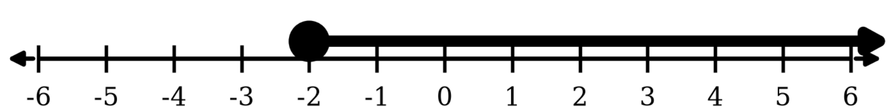
1. Solve for x. $\frac{3}{4} = \frac{x}{20}$ ________________

2. Simplify. $-7 - (-12) \div 3$ ________________

3. Write $\sqrt{72}$ in simplest radical form. ________________

4. Simplify, no negative exponents. $\left(3x^2y^{-3}\right)^4$ ________________

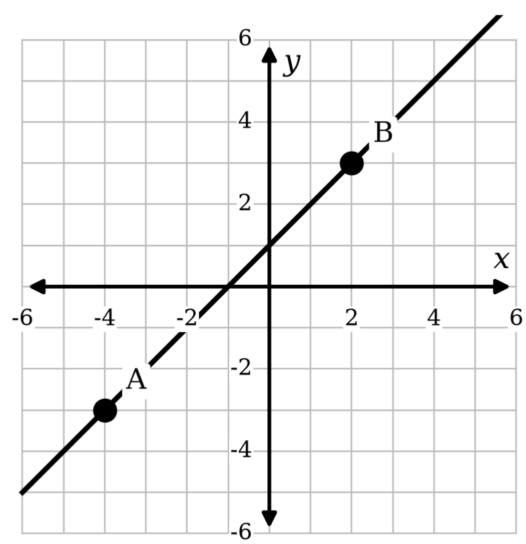
5. Graph the solution set of $x \geq -2$ on the number line below.



6. Find the slope of the line through $A(-4, -3)$ and $B(2, 3)$, then write its equation in slope–intercept form.

$m =$ ________________

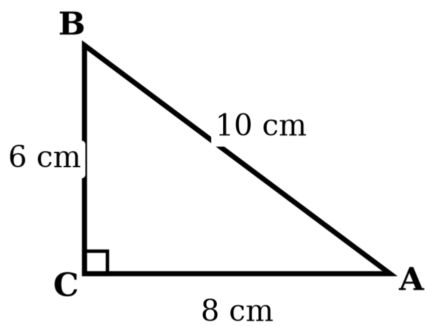
$y =$ ________________



Geometry

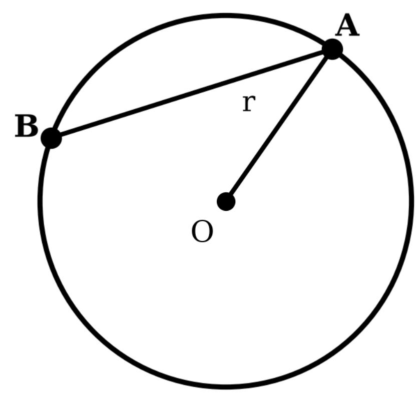
FIGURES DRAWN FROM THE NUMBERS, NOT FROM A PROMPT

7. $\triangle ABC$ has a right angle at C. Find $\overline{AB}$.



$\overline{AB} =$ ____________________

8. $\overline{OA}$ is a radius and $\overline{AB}$ a chord. If $m\angle AOB = 105°$, find the measure of minor arc AB.



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9. Find the volume of the cylinder. Leave your answer in terms of π.

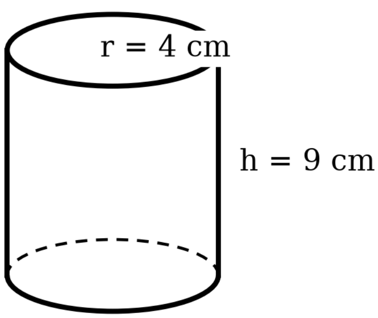
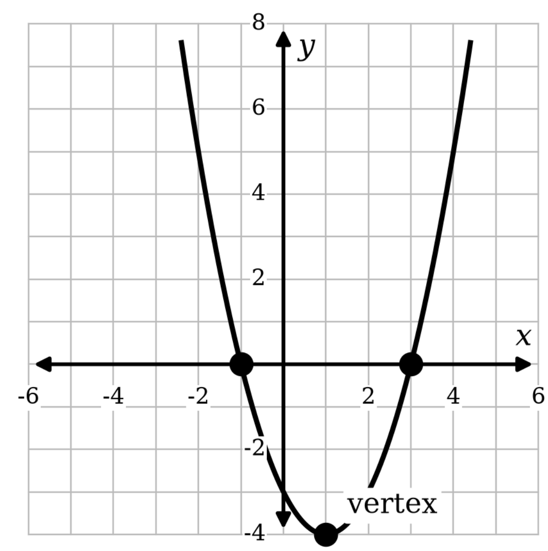


Figure not drawn to scale.

$V =$ ____________________

10. The graph of $y = x^2 - 2x - 3$ is shown. Name the zeros and the vertex.



zeros ____________________

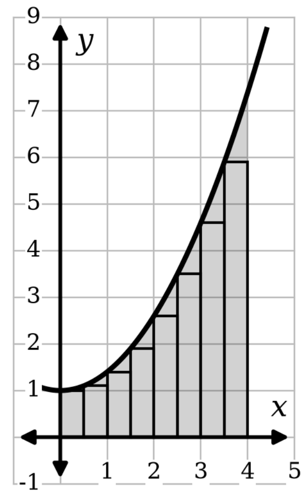
Calculus & Data

THE SAME ENGINE, FOUR GRADES FURTHER UP

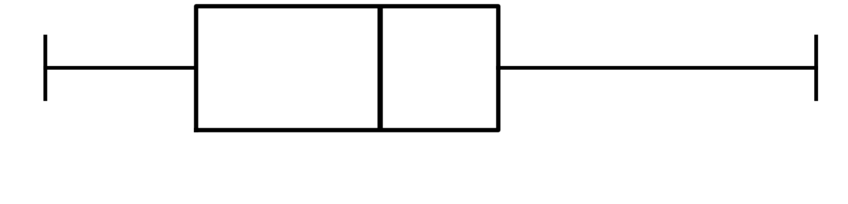
11. Estimate $\int_0^4 \left(0.4x^2 + 1\right)\,dx$ with a left Riemann sum using $n = 8$ rectangles, then find the exact value.

$L_8 \approx$ ________________

exact ________________



12. Find the five-number summary for the reading-minutes data.



min ________________ Q_1 ________________ median ________________ Q_3 ________________ max ________________

13. Differentiate. $\frac{d}{dx}\left[x^3 \sin x\right]$ ________________

14. Integrate by parts. $\int xe^x\,dx$ ________________

The Blackboard

THE FAR END OF THE LADDER — CAN IT SET THIS?

Every line below came out of the same renderer as the fractions on sheet one. No images, no pasted screenshots — real text in real fonts, embedded in the PDF.

Analysis

$$\forall \varepsilon > 0\ \exists \delta > 0 :\ 0 < |x-a| < \delta \implies |f(x) - L| < \varepsilon$$

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\, dx \qquad \zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$$

Linear algebra & algebraic structure

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

$$\mathbb{R}^n, \quad \mathbb{Z}/n\mathbb{Z}, \quad \mathfrak{su}(2), \quad \mathcal{C} \xrightarrow{F} \mathcal{D}$$

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle g}\big\downarrow & & \big\downarrow{\scriptstyle h} \\ C & \xrightarrow[k]{} & D \end{array}$$

Physics

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \hat{H} \Psi(\mathbf{r}, t) \qquad \langle \psi | \hat{A} | \psi \rangle$$

$$R_{\mu\nu} - \tfrac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

$$\mathcal{L} = -\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}\, (i\gamma^\mu D_\mu - m)\, \psi \qquad (i\partial\!\!\!/ - m)\psi = 0$$

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2} g^{\lambda\sigma} \left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu} \right)$$

Worked, step by step — the same machinery a 7th grader gets

$$\begin{aligned} 3x + 2 &= 17 \\ 3x &= 15 \\ x &= 5 \end{aligned}$$

Answer Key

EVERY LINE BELOW WAS COMPUTED, NOT TYPED

These answers were produced by a computer algebra system from the same expressions the worksheet printed. Nobody wrote them down and nobody checked them by hand, which is exactly why they are trustworthy.

1. **P1** $\frac{3}{4} = \frac{x}{20} \rightarrow 15$ (15.0000)
2. **P2** $-7 - (-12) \div 3 \rightarrow -3$ (-3.0000)
3. **P3** $\sqrt{72} \rightarrow 6\sqrt{2}$ (8.4853)
4. **P4** $\left(3x^2y^{-3}\right)^4 \rightarrow \frac{81x^8}{y^{12}}$
5. **P5** $2x^2 - 7x - 15 = 0 \rightarrow -\frac{3}{2},\ 5$
6. **P6** $\log_3\left(\frac{x^2}{9}\right) = 4 \rightarrow -27,\ 27$
7. **P7** $\frac{d}{dx}\left[x^3 \sin x\right] \rightarrow x^2\left(x\cos\left(x\right) + 3\sin\left(x\right)\right)$
8. **P8** $\int_0^\pi \sin x\, dx \rightarrow 2$ (2.0000)
9. **P9** $\int xe^x\, dx \rightarrow (x-1)\, e^x + C$
10. **P10** $\lim_{x\to 0} \frac{\sin x}{x} \rightarrow 1$ (1.0000)
11. **P11** $\sum_{k=1}^n k^2 \rightarrow \frac{n(n+1)(2n+1)}{6}$
12. **P12** hypotenuse, legs $8, 6 \rightarrow 10$ (10.0000)
13. **P13** $V = \pi r^2 h,\ r = 4,\ h = 9 \rightarrow 144\pi$ (452.3893)

Figure audit

1. **PASS** triangle side labels match the drawing labels 8,10,6 vs measured 8.000,10.000,6.000
2. **PASS** the right angle really measures 90 angles 90.00, 36.87, 53.13
3. **PASS** interior angles sum to 180 180
4. **PASS** Pythagorean answer equals the drawn hypotenuse key 10 vs drawn 10
5. **PASS** plotted line has the slope the key claims (m = 1) m = 1
6. **PASS** cylinder volume key matches its printed labels key 452.3893 vs pi*4^2*9 = 452.3893
7. **PASS** cylinder figure is declared NOT TO SCALE drawn r=1.5 h=3.2, labelled r=4 h=9 — sheet must print "not to scale"
8. **PASS** box plot median is computed, not eyeballed {"path":"C:\\Users\\Valued Customer\\OneDrive\\Desktop\\WEBSITES\\Jumpdrive\\NCHO\\workbooks\\math-lab\\out\\fig\\boxplot.png","bytes":4220,"min":12,"q1":16.5,"median":22,"q3":25.5,"max":35,"iqr":9,"collisions":[],"clipped":[],"dpi":300,"px":[1380,450],"inches":[4.6,1.5]}
9. **PASS** every problem's TeX parses 13/13